Fiscal Year:	FY 2006	**Task Last Updated:**	FY 10/10/2007
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Project Title:	Optimization of Cryogenic Cooling of Protein Crystals		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Bio-nanotechnology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	43606-3390	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-03
Start Date:	06/01/2001	**End Date:**	11/30/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	3
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
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Grant/Contract No.:	NAG 8-1838		
Performance Goal No.:			
Performance Goal Text:			
	Structure determination of large proteins, viruses and protein complexes is accomplished using X-ray diffraction techniques. Protein crystallization is the essential first step in X-ray crystallographic structure determination of biological macromolecules. For these studies, protein crystal quality and crystallinity are very important. There is evidence that protein crystals grown in microgravity have reduced mosaicity and are larger in size than terrestrially grown crystals. In order to maintain the protein crystal quality, structure and composition of both microgravity grown and terrestrially grown crystals in ionizing X-ray radiation, biological crystals are cooled to cryogenic temperatures. Cryogenic cooling of biological crystals has led to significantly longer lifetimes of crystals exposed to the ionizing radiation of X-rays. This improvement of crystal lifetime is largely attributed to the reduced diffusion of free radicals formed in the ionizing beam. Optimization of the flash cooling process will prevent loss of biological crystals during		

Task Description:	cooling. This is especially important for biological molecules that are available in limited quantities. Protein crystals contain a significant fraction of solvent in the large channels within the crystal. This solvent is primarily disordered water. In bringing the crystal to low temperatures, this solvent must be vitrified without the formation of ice, which would expand within the solvent channels, destroying the protein crystal structure and introducing additional Bragg scattering. In general, amorphous (vitreous) solids are formed with rapid cooling. Most aqueous solutions require the addition of chemicals termed cryoprotectants in order to form glasses at practically attainable cooling rates. Thus, cryoprotectants such as glycerol, d-sorbitol, 1-2- propanediol, dimethyl sulfoxide or cryosalts are used to raise the glass transition temperature and depress the melting point. The cooling rates required for vitrification are then attainable using liquid or gaseous cryogens such as nitrogen or helium. In order to formulate cryoprotectant solutions and cryocooling techniques, models of the heat transfer process and property data or models for both cryoprotectant solutions and crystals is needed. Predictive models of thermal properties and heat transfer will allow prediction of flash cooling protocols for a variety of crystal sizes and geometries, reducing the loss of crystals due to trial and error approaches. The overall goal of this project is to develop predictive heat transfer models and property correlations for cooling protein crystals in helium and nitrogen gas streams. The overall goal of this project is to optimize flash cooling protocols for protein crystals. In order to generalize these protocols for crystals of varying size, using either nitrogen or helium streams at varying temperatures a predictive heat transfer model will be developed. The crystal diffraction quality was then assessed for crystals subjected to varying cooling protocols and data collected at varying temperature. The research focused on the the specific objectives listed below. 1. Measure thermal and physical properties of cryoprotectant solutions and correlate the glass transition to composition for multi-component mixtures. 2. Develop a predictive model for convective and conductive heat transfer for an open flow helium and nitrogen cryostat. 3. Analyze diffraction data collected at varying cryogenic temperatures to assess the effect of collection temperature on data quality.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	X-ray diffraction techniques are applied to crystals of biological molecules and are used to determine the atomic level structure of biological molecules. Structure of biological molecules is related to their function and knowledge of structure at the atomic level leads to a better understanding of molecules that are essential to homeostasis and disease processes. The X-ray diffraction techniques used to obtain structural information can damage biological molecule crystals due to the ionizing radiation of the X-ray beam. This damage is greatly reduced if the crystals are first rapidly cooled (flash cooled) to cryogenic temperatures. The flash cooling process itself can damage crystals and render them unusable for X-ray diffraction studies. In this project we have developed models of the flash cooling process and property prediction to ascertain cryoprotectant conditions for protein crystals systems. Use of these models allows judicious choice of cryo-cooling conditions and reduces crystal loss, enabling more rapid and successful X-ray data collection. Cooling biological crystals to cryogenic temperatures greatly reduces radiation damage during X-ray crystallographic structure determination. Very high quality crystals are needed for high resolution or atomic level structures. This structural information is used in pharmaceutical, biotechnology, agricultural and medical research. The results of this research allows crystallographers to select cryoprotectant conditions based on the properties of their crystal system. This allows effective flash cooling crystals of biological molecules, minimizing trial and error tests. Optimization of the flash cooling and data collection process will help prevent loss of crystals during cryogenic data collection.
	1. Measure thermal and physical properties of cryoprotectant solutions and correlate the glass transition to composition for multi-component mixtures. Vitrification is achieved by rapidly cooling protein crystals and the surrounding solution or mother liquor from room temperature to a temperature below the glass transition of the crystal and solution. Because of the high volume fraction of aqueous solution in the protein crystalchannels (~20 to 75%), this solution often behaves as bulk solution. Solution composition ranges from pure water to multi-components aqueous solutions of variable composition. Pure water has a reported glass transition temperature between 130 K to 140 K. It is not possible to achieve complete vitrification of pure water (or many of the aqueous solutions in the solvent channels) for the sample size representative of protein crystals, as this would require unattainably high cooling rates. Thus, cryoprotectants such as glycerol, d-sorbitol, 1,2 propanediol, poly(ethylene glycol), dimethyl sulfoxide or salts are used to raise the glass transition temperature and depress the melting and homogenous ice nucleation temperature, thus making it practically feasible to vitrify samples with commercially available cryogens and cryostats. Glycerol and ethylene glycol are often used as cryoprotectants. The glass transition temperature, devitrification (or recrystallization) temperature and melting temperature for aqueous binary mixtures of glycerol, ethylene glycol and lithium chloride and some ternary mixtures of glycerol-salt (sodium chloride or ammonium sulfate)-water were measured. Transition temperatures of a selected set of Hampton Screen I (Hampton Research, HR2-100), a widely used screen for crystallization of macromolecules, and glycerol mixtures were measured. Transition temperatures for buffer-salt-glycerol mixtures used for D-xylose isomerase crystallization were also measured. Deuterated solutions are largely used in neutron diffraction studies. However, they may also serve as cryoprotectants in X-ray diffraction. High power X-ray beams at synchroton sources may produce diffraction spots of deuterium (an isotope of hydrogen). The cryoprotective behaviour of deuterated glycerol-d8 and ethylene glycol-d6 solutions made in heavy water (D2O) were also measured to give an estimate of cryoprotective behavior of deuterated solutions. Solutions representative of those used in protein crystallization studies are multi-component mixtures of water, buffer salts, precipitants, other additives and cryoprotectants such a glycerol. There are limited data for Tg of multicomponent mixtures and few studies comparing experimental and predicted values. The glass transition temperature gives information about the target temperature to be reached in order to avoid ice formation. Knowledge of the compositional

	dependency of Tg for the multi-component solutions used in protein crystallization can allow selection of cryoprotectant concentration based on a target value of Tg. A differential scanning calorimeter (DSC) was used to measure the glass transition (Tg), devitrification (Td) and melting (Tm) temperatures of binary and multi-component solutions used in protein crystallization. 2. Develop a predictive model for convective and conductive heat transfer for an open flow helium and nitrogen cryostat.
Task Progress:	In order to assess the impact of sample size on flash cooling, a series of Hampton Screen I crystallization solutions were flash cooled to ascertain the minimum glycerol concentration needed for vitrification of solutions in loops of nominal sizes of 1.0, 0.5 and 0.1 mm. A 100 K nitrogen gas stream was generated using an Oxford Cryostream series 600 nitrogen cryostat. Glycerol concentration was varied in 5 % (v/v) increments. Samples were considered to be vitrified when the ratio of sample intensity to background was less than 1.35. The minimum glycerol concentration required for vitrification of the Hampton screen I solutions obtained from the X-ray data analysis is listed in table 3. The glycerol concentration needed for vitrification was found to be dramatically lower for the smaller loop sizes as seen in table 1. Indeed, if the sample is small enough, no cryoprotectant is required. In order to vitrify an aqueous solution, cooling must be rapid enough such that ice nucleation does not occur during the time period that the solution is between its melting temperature and the glass transition. Depressing melting temperature and increasing the glass transition temperature of the solution for a given cooling rate decreases the time that the sample is between these temperatures, leading to a decreased likelihood of ice formation. Cryoprotectants, such as glycerol, raise the glass transition temperature (and depress the melting point) and increase the likelihood of successful vitrification. Additionally, many cryoprotectants increase solution viscosity, which also slows ice nucleation rates. Samples in larger cryoloops cool at a slower rate than those in smaller loops. In order to maintain the same time interval between solution melting and the glass transition temperatures for large samples compared to small samples, glass transition temperatures must be higher for larger samples. dependent thermodynamic and transport properties are known. With the knowledge of heat capacities and densities of solutions for a large range of glycerol concentrations, the glass transition temperatures (and therefore glycerol concentrations) required to successfully vitrify an unknown sample may be predicted with a selected critical vitrification time. 3. Analyze diffraction data collected at varying cryogenic temperatures to assess the effect of collection temperature on data quality. In experiments conducted at the Bio-CARS beamline 14 BM-C (APS, Argonne National Laboratory, USA), Streptomyces rubiginosus D-xylose isomerase (EC5.3.1.5) crystals were used to test the effect of cryogen temperature on radiation damage. Crystals cooled using a helium cryostat at a 8 K set temperature consistently showed less decay in the signal-to-noise ratio and in average intensity compared to those cooled with a nitrogen cryostat set to 100 K. Multiple crystals grown using ammonium sulphate as precipitant were used at each cryostat set temperature and comparisons were made for crystals of similar size and diffraction resolution. Maximum resolution for the crystals was 1.1 to 1.3 Å, with He at < 20K extending the lifetime of the high resolution data >25% compared to crystals cooled with N2 at 100 K.

Bibliography Type:	Description: (Last Updated: 10/11/2007)
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